U-ADMISSIBLE BLOWUPS SEPARATE DISJOINT CLOSED SUBSCHEMES AND DISJOINT CONSTRUCTIBLE SETS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3239-3253 of sources/stacks-spaces-pushouts/pinned-original-source.tex; proof span is lines 3255-3294, the complete author proof (38 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 13 of queue_spaces-pushouts.json (pb=1882, stmt 3239-3253, proof 3255-3294).

Editorial note. spaces-pushouts wave-1: fresh adjudication accept (2026-10-14); six-check dispatched pool (§2 row appended 02:4x).

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-separate-disjoint-locally-closed-by-blowing-up' also exists in stacks-flat — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 086B: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3294–3294. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 086F: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3294–3294. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0864: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3294–3294. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0853: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3294–3294. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-pushouts.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0855: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3294–3294. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 084Q: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3294–3294. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03XQ: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3294–3294. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let X be a quasi-compact and quasi-separated algebraic space over S. Let $U \subset X$ be a quasi-compact open.*

1. *If $Z_1, Z_2 \subset X$ are closed subspaces of finite presentation such that $Z_1 \cap Z_2 \cap U = \emptyset$, then there exists a U-admissible blowing up $X' \to X$ such that the strict transforms of Z_1 and Z_2 are disjoint.*
2. *If $T_1, T_2 \subset |U|$ are disjoint constructible closed subsets, then there is a U-admissible blowing up $X' \to X$ such that the closures of T_1 and T_2 are disjoint.*

Proof. Proof of (1). The assumption that $Z_i \to X$ is of finite presentation signifies that the quasi-coherent ideal sheaf $\mathcal{I}_i$ of Z_i is of finite type, see Morphisms of Spaces, Lemma 084Q. Denote $Z \subset X$ the closed subspace cut out by the product $\mathcal{I}_1\mathcal{I}_2$. Observe that $Z \cap U$ is the disjoint union of $Z_1 \cap U$ and $Z_2 \cap U$. By Divisors on Spaces, Lemma 086F there is a $U \cap Z$-admissible blowup $Z' \to Z$ such that the strict transforms of Z_1 and Z_2 are disjoint. Denote $Y \subset Z$ the center of this blowing up. Then $Y \to X$ is a closed immersion of finite presentation as the composition of $Y \to Z$ and $Z \to X$ (Divisors on Spaces, Definition 086B and Morphisms of Spaces, Lemma 03XQ). Thus the blowing up $X' \to X$ of Y is a U-admissible blowing up. By general properties of strict transforms, the strict transform of Z_1, Z_2 with respect to $X' \to X$ is the same as the strict transform of Z_1, Z_2 with respect to $Z' \to Z$, see Divisors on Spaces, Lemma 0864. Thus (1) is proved.

Proof of (2). By Limits of Spaces, Lemma 0855 there exists a finite type quasi-coherent sheaf of ideals $\mathcal{J}_i \subset \mathcal{O}_U$ such that $T_i = V(\mathcal{J}_i)$ (set theoretically). By Limits of Spaces, Lemma 0853 there exists a finite type quasi-coherent sheaf of ideals $\mathcal{I}_i \subset \mathcal{O}_X$ whose restriction to U is $\mathcal{J}_i$. Apply the result of part (1) to the closed subspaces $Z_i = V(\mathcal{I}_i)$ to conclude. □